Lemma：$\mathbb{R}^2$ 是 $\mathbb{R}$一向量空间，$f:\mathbb{R}^2\to\mathbb{R}^2$ 是一个任意的映射，则有：
下列命题等价：

① $f:\mathbb{R}^2\to\mathbb{R}^2$ 是线性映射

② 存在唯一的 $A\in M_{2\times2}(\mathbb{R})$，使得对 $\forall\, x\in\mathbb{R}^2$，都有 $f(x)=Ax$.

Proof：①⇒② 已证.

②⇒①：$\because$ 存在唯一的 $A\in M_{2\times2}(\mathbb{R})$，使得对 $\forall\, x\in\mathbb{R}^2$，都有
$f(x)=Ax$

$\therefore A\in M_{2\times2}(\mathbb{R})$，且对 $\forall\, x\in\mathbb{R}^2$，有：$f(x)=Ax$.

对 $\forall$ ~~…~~ $v_1, v_2\in\mathbb{R}^2$，有：

$$f(v_1+v_2)=A(v_1+v_2)=Av_1+Av_2=f(v_1)+f(v_2)$$

对 $\forall\, t\in\mathbb{R}$，$\forall\, v\in\mathbb{R}^2$，有：

$$f(tv)=A(tv)=t(Av)=tf(v)$$

$\therefore f:\mathbb{R}^2\to\mathbb{R}^2$ 是线性映射. $\square$

Lemma：对 $\forall\, A\in M_{2\times2}(\mathbb{R})$，设 $A=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$. 设
$\alpha_1=(a\ \ b)$，$\alpha_2=(c\ \ d)$，$\beta_1=\begin{pmatrix}a\\ c\end{pmatrix}$ $\beta_2=\begin{pmatrix}b\\ d\end{pmatrix}$

则有：下列条件等价：

① A是 2×2 实正交矩阵

② $\|\alpha_1\|=\|\alpha_2\|=1$ 且 $\alpha_1\perp\alpha_2$

③ $\|\beta_1\|=\|\beta_2\|=1$ 且 $\beta_1\perp\beta_2$.

Proof: $\because A=\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2\times2}(\mathbb{R}) \quad \therefore {}^tA=\begin{pmatrix} a & c \\ b & d \end{pmatrix} \in M_{2\times2}(\mathbb{R})$

$$\therefore {}^tA\cdot A=\begin{pmatrix} a & c \\ b & d \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}=\begin{pmatrix} a^2+c^2 & ab+cd \\ ab+cd & b^2+d^2 \end{pmatrix}$$

$$A\cdot {}^tA=\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix}=\begin{pmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{pmatrix}$$

对 $\forall\, x=\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\in\mathbb{R}^2$, 有: $(x|x)=x_1^2+x_2^2\in\mathbb{R}_{\geqslant 0}$

$\therefore \|x\|=\sqrt{(x|x)}=\sqrt{x_1^2+x_2^2}\in\mathbb{R}_{\geqslant 0}$

$\therefore ① \Leftrightarrow A\cdot{}^tA=I_{2\times2} \Leftrightarrow \begin{pmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{pmatrix}=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Leftrightarrow a^2+b^2=1$ 且 $ac+bd=0$ 且 $c^2+d^2=1$

$\Leftrightarrow (\alpha_1|\alpha_1)=1$ 且 $(\alpha_2|\alpha_2)=1$ 且 $(\alpha_1|\alpha_2)=0$

$\Leftrightarrow \|\alpha_1\|=1$ 且 $\|\alpha_2\|=1$ 且 $\alpha_1\perp\alpha_2 \Leftrightarrow ②$

$① \Leftrightarrow {}^tA\cdot A=I_{2\times2} \Leftrightarrow \begin{pmatrix} a^2+c^2 & ab+cd \\ ab+cd & b^2+d^2 \end{pmatrix}=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Leftrightarrow a^2+c^2=1$ 且 $b^2+d^2=1$ 且 $ab+cd=0$

$\Leftrightarrow (\beta_1|\beta_1)=1$ 且 $(\beta_2|\beta_2)=1$ 且 $(\beta_1|\beta_2)=0$

$\Leftrightarrow \|\beta_1\|=1$ 且 $\|\beta_2\|=1$ 且 $\beta_1\perp\beta_2 \Leftrightarrow ③$ $\square$

定理（$\mathbb{R}^2$上的保距映射）$\mathbb{R}^2$是$\mathbb{R}$一向量空间.

对$\forall\, x=\begin{pmatrix}x_1\\x_2\end{pmatrix}\in\mathbb{R}^2$，$\forall\, y=\begin{pmatrix}y_1\\y_2\end{pmatrix}\in\mathbb{R}^2$，定义$x$和$y$的标准内积

为 $(x|y)=x_1y_1+x_2y_2$

对$\forall\, x\in\mathbb{R}^2$，定义$x$的Euclid长度为$\|x\|=\sqrt{(x|x)}\in\mathbb{R}_{\geq 0}$

$f:\mathbb{R}^2\longrightarrow\mathbb{R}^2$是一个任意的映射. 则下列命题等价：

① 对$\forall\, x,y\in\mathbb{R}^2$，有：$\|f(x)-f(y)\|=\|x-y\|$

② 存在唯一的2×2实正交矩阵A和唯一的$b\in\mathbb{R}^2$，使得对$\forall\, x\in\mathbb{R}^2$，都有$f(x)=Ax+b$.

Proof：①$\Rightarrow$②：$\because f:\mathbb{R}^2\longrightarrow\mathbb{R}^2$是一个映射 $\therefore f\left(\begin{pmatrix}0\\0\end{pmatrix}\right)\in\mathbb{R}^2$

令$b=f\left(\begin{pmatrix}0\\0\end{pmatrix}\right)$ $\therefore b\in\mathbb{R}^2$.

定义映射 $g:\mathbb{R}^2\longrightarrow\mathbb{R}^2$ $\therefore g:\mathbb{R}^2\to\mathbb{R}^2$是一个映射，

$x\longmapsto f(x)-b$.

且有 $g\left(\begin{pmatrix}0\\0\end{pmatrix}\right)=f\left(\begin{pmatrix}0\\0\end{pmatrix}\right)-b=f\left(\begin{pmatrix}0\\0\end{pmatrix}\right)-f\left(\begin{pmatrix}0\\0\end{pmatrix}\right)=\begin{pmatrix}0\\0\end{pmatrix}$

对$\forall\, x,y\in\mathbb{R}^2$，有：$\|g(x)-g(y)\|=\|(f(x)-b)-(f(y)-b)\|$

$=\|f(x)-b-f(y)+b\|=\|f(x)-f(y)\|=\|x-y\|$

对$\forall\, x\in\mathbb{R}^2$，有：$\|g(x)\|=\left\|g(x)-\begin{pmatrix}0\\0\end{pmatrix}\right\|=\left\|g(x)-g\left(\begin{pmatrix}0\\0\end{pmatrix}\right)\right\|$

$=\left\|x-\begin{pmatrix}0\\0\end{pmatrix}\right\|=\|x\|$

对 $\forall x, y \in \mathbb{R}^2$，有：$g(x), g(y) \in \mathbb{R}^2$.

$$\therefore (g(x) \mid g(y)) = \frac{1}{2}\left(\|g(x)\|^2 + \|g(y)\|^2 - \|g(x)-g(y)\|^2\right)$$

$$= \frac{1}{2}\left(\|x\|^2 + \|y\|^2 - \|x-y\|^2\right) = (x \mid y)$$

对 $\forall x, y \in \mathbb{R}^2$，有：

$$(g(x+y)-g(x)-g(y) \mid g(x+y)-g(x)-g(y))$$

$$= (g(x+y) \mid g(x+y)) - (g(x+y) \mid g(x)) - (g(x+y) \mid g(y))$$
$$- (g(x) \mid g(x+y)) + (g(x) \mid g(x)) + (g(x) \mid g(y))$$
$$- (g(y) \mid g(x+y)) + (g(y) \mid g(x)) + (g(y) \mid g(y))$$

$$= (x+y \mid x+y) - (x+y \mid x) - (x+y \mid y) - (x+y \mid x) - (x+y \mid y)$$
$$+ (x \mid x) + 2(x \mid y) + (y \mid y)$$

$$= (x \mid x) + 2(x \mid y) + (y \mid y) - (x \mid x) - (x \mid y) - (x \mid y) - (y \mid y)$$
$$- (x \mid x) - (x \mid y) - (x \mid y) - (y \mid y) + (x \mid x) + 2(x \mid y) + (y \mid y)$$

$$= 0$$

$$\therefore g(x+y) - g(x) - g(y) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \therefore g(x+y) = g(x) + g(y)$$

对 $\forall \lambda \in \mathbb{R}$, $\forall x \in \mathbb{R}^2$，有：

$$\cancel{g(\lambda x) - \lambda} \qquad (g(\lambda x) - \lambda g(x) \mid g(\lambda x) - \lambda g(x))$$

$$= (g(\lambda x) \mid g(\lambda x)) - (g(\lambda x) \mid \lambda g(x)) - (\lambda g(x) \mid g(\lambda x)) + (\lambda g(x) \mid \lambda g(x))$$

$= (\lambda x|\lambda x) - \lambda(\lambda x|x) - \lambda(\lambda x|x) + \lambda^2(x|x)$

$= \lambda^2(x|x) - \lambda^2(x|x) - \lambda^2(x|x) + \lambda^2(x|x) = 0$

$\therefore g(\lambda x) - \lambda g(x) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\quad\therefore g(\lambda x) = \lambda g(x)$

$\therefore g: \mathbb{R}^2 \to \mathbb{R}^2$是一个线性映射.

$\therefore$存在唯一的$A \in M_{2\times 2}(\mathbb{R})$，使得对$\forall x \in \mathbb{R}^2$，都有$g(x) = Ax$

设$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \in M_{2\times 2}(\mathbb{R})$

对于$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathbb{R}^2$，有：$\left\| g\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \right\| = \left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\|$

$\therefore \left\| \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} \right\| = \left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\| = 1$

对于$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{R}^2$，有：$\left\| g\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) \right\| = \left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\|$

$\therefore \left\| \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} \right\| = \left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\| = 1$

对于$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathbb{R}^2$，$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{R}^2$，有：$\left(g\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \middle| g\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) \right) = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \middle| \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$

$\therefore \left(\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} \middle| \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} \right) = 0$ $\quad\therefore \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} \perp \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$

$\therefore A$是2×2实正交矩阵.

$\therefore$对$\forall x \in \mathbb{R}^2$，有：$g(x) = Ax = f(x) - b$ $\quad\therefore f(x) = Ax + b$

$\therefore$存在2×2实正交矩阵A和$b \in \mathbb{R}^2$，使得对$\forall x \in \mathbb{R}^2$，有：

$f(x) = Ax + b$. 存在性得证.

假设存在 2×2 实正交矩阵 A 和 $b\in\mathbb{R}^2$，s.t. 对 $\forall x\in\mathbb{R}^2$，都有

$f(x)=Ax+b$.

假设还存在 2×2 实正交矩阵 C 和 $d\in\mathbb{R}^2$，s.t. 对 $\forall x\in\mathbb{R}^2$，都有

~~$f(x)=Cx+d$~~ $f(x)=Cx+d$

$\because \begin{pmatrix}0\\0\end{pmatrix}\in\mathbb{R}^2$，且 $f\left(\begin{pmatrix}0\\0\end{pmatrix}\right)=A\begin{pmatrix}0\\0\end{pmatrix}+b$，$f\left(\begin{pmatrix}0\\0\end{pmatrix}\right)=C\begin{pmatrix}0\\0\end{pmatrix}+d$

$\therefore \begin{pmatrix}0\\0\end{pmatrix}+b=\begin{pmatrix}0\\0\end{pmatrix}+d$ $\quad\therefore b=d$.

设 $A=\begin{pmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{pmatrix}$ $\quad C=\begin{pmatrix}c_{11}&c_{12}\\c_{21}&c_{22}\end{pmatrix}$

$\therefore$ 对于 $\begin{pmatrix}1\\0\end{pmatrix}\in\mathbb{R}^2$，有：$f\left(\begin{pmatrix}1\\0\end{pmatrix}\right)=A\begin{pmatrix}1\\0\end{pmatrix}+b=C\begin{pmatrix}1\\0\end{pmatrix}+b$

$\therefore \begin{pmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}c_{11}&c_{12}\\c_{21}&c_{22}\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}$ $\quad\therefore \begin{pmatrix}a_{11}\\a_{21}\end{pmatrix}=\begin{pmatrix}c_{11}\\c_{21}\end{pmatrix}$

$\therefore a_{11}=c_{11}$ 且 $a_{21}=c_{21}$

对于 $\begin{pmatrix}0\\1\end{pmatrix}\in\mathbb{R}^2$，有：$f\left(\begin{pmatrix}0\\1\end{pmatrix}\right)=A\begin{pmatrix}0\\1\end{pmatrix}+b=C\begin{pmatrix}0\\1\end{pmatrix}+b$

$\therefore \begin{pmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}c_{11}&c_{12}\\c_{21}&c_{22}\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix}$ $\quad\therefore \begin{pmatrix}a_{12}\\a_{22}\end{pmatrix}=\begin{pmatrix}c_{12}\\c_{22}\end{pmatrix}$

$\therefore a_{12}=c_{12}$ 且 $a_{22}=c_{22}$ $\quad\therefore A=C$ 且 $b=d$ 唯一性得证.

②$\Rightarrow$①：$\because$ 存在唯一的 2×2 实正交矩阵 A 和唯一的 $b\in\mathbb{R}^2$，s.t.

对 $\forall x\in\mathbb{R}^2$，都有 $f(x)=Ax+b$

设 $A=\begin{pmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{pmatrix}$. $\quad$ 对 $\forall x,y\in\mathbb{R}^2$，设 $x=\begin{pmatrix}x_1\\x_2\end{pmatrix}$，$y=\begin{pmatrix}y_1\\y_2\end{pmatrix}$

$$\therefore \|f(x)-f(y)\| = \|(Ax+b)-(Ay+b)\| = \|Ax+b-Ay-b\|$$

$$= \|Ax-Ay\| = \|A(x-y)\| = \left\| \begin{pmatrix} a_{11}(x_1-y_1)+a_{12}(x_2-y_2) \\ a_{21}(x_1-y_1)+a_{22}(x_2-y_2) \end{pmatrix} \right\|$$

$$= \sqrt{\left(\begin{pmatrix} a_{11}(x_1-y_1)+a_{12}(x_2-y_2) \\ a_{21}(x_1-y_1)+a_{22}(x_2-y_2) \end{pmatrix} \middle| \begin{pmatrix} a_{11}(x_1-y_1)+a_{12}(x_2-y_2) \\ a_{21}(x_1-y_1)+a_{22}(x_2-y_2) \end{pmatrix} \right)}$$

$$= \sqrt{\left(a_{11}(x_1-y_1)+a_{12}(x_2-y_2)\right)^2 + \left(a_{21}(x_1-y_1)+a_{22}(x_2-y_2)\right)^2}$$

$$= \sqrt{(a_{11}^2+a_{21}^2)(x_1-y_1)^2 + 2(a_{11}a_{12}+a_{21}a_{22})(x_1-y_1)(x_2-y_2) + (a_{12}^2+a_{22}^2)(x_2-y_2)^2}$$

$$= \sqrt{(x_1-y_1)^2+(x_2-y_2)^2} = \|x-y\| \qquad \square$$